GLOBAL CYBERSECURITY THREAT INTELLIGENCE

COMPREHENSIVE CASE STUDY & ANALYSIS REPORT

MBAG – 406 – Visual Analytics & Story Telling Project

S Adithya

24030302038

II MBA

Department of Management & Commerce

Sri Sathya Sai Institute of Higher Learning

Brindavan Campus

Table of Contents

EXECUTIVE SUMMARY 3
DATASET OVERVIEW & METHODOLOGY 3
FINANCIAL LOSS & INCIDENT TREND ANALYSIS 4
ATTACK VECTOR DISTRIBUTION 6
TARGET INDUSTRY VULNERABILITY ANALYSIS 7
DEFENCE MECHANISM ADOPTION 7
INCIDENT RESOLUTION TIME ANALYSIS 8
STRATEGIC RECOMMENDATIONS 9
CONCLUSION 10

EXECUTIVE SUMMARY

This case study presents a comprehensive analysis of global cybersecurity threats recorded between 2015 and 2024, drawing on a dataset of 3,000 verified incidents spanning 10 countries and 7 major industry sectors. The analysis was conducted using the Power BI dashboard — Global Cybersecurity Threat Intelligence — which synthesises multi-dimensional threat data to surface patterns in attack behaviour, financial impact, defence efficacy, and incident resolution. The findings reveal an alarming, sustained escalation in both the frequency and financial severity of cyber incidents globally. Cumulatively, the 3,000 incidents examined resulted in a total financial loss of approximately $151.48 billion USD, impacted over 1.51 billion users, and demanded an average resolution time of 36.48 hours per incident — underscoring the systemic fragility of current cybersecurity postures across industries and geographies.

Metric	Value
Total Incidents	3,000 (2015–2024)
Users Affected	1.51 Bn (Global)
Avg Resolution Time	36.48 Hrs (Per Incident)
Total Financial Loss (USD)	$151.48 Bn (Cumulative)

Key themes emerging from the analysis include: the dominance of DDoS and Phishing as primary attack vectors; significant concentration of financial losses in the IT and Banking sectors; broadly uniform geographic exposure, with the UK recording the highest cumulative losses; and a striking under-utilisation of AI-based detection systems despite their proven effectiveness.

KEY INSIGHT: No single defence mechanism has achieved dominant adoption — antivirus tools lead narrowly with only 20.9% frequency, signalling an urgent need for layered, AI-augmented security frameworks.

DATASET OVERVIEW & METHODOLOGY

The dataset underpinning this analysis comprises 3,000 cybersecurity incident records collected across a decade (2015–2024). Each record captures ten attributes covering geographic origin, temporal context, attack typology, targeted sector, financial impact, user exposure, threat actor classification, exploited vulnerability, defence mechanism deployed, and resolution duration

Attribute	Description	Coverage
Country	The nation where the incident occurred	10 countries
Year	Year of incident	2015 – 2024
Attack Type	Primary attack vector	6 categories
Target Industry	Affected sector	7 industries
Financial Loss	Monetary impact (USD Millions)	$0.01M – $99.99M
Users Affected	Number of impacted individuals	Up to 1M+
Attack Source	Origin of the threat actor	4 categories
Security Vulnerability	Exploited weakness	4 categories
Defence Mechanism	Deployed countermeasure	5 types
Resolution Time	Hours to full containment	1 – 72 hours

The analysis follows a structured business intelligence methodology: raw data was ingested into Power BI, cleaned and normalised, then visualised across six dashboard panels covering temporal trends, geospatial exposure, attack typology, industry vulnerability, defence adoption, and resolution efficiency.

FINANCIAL LOSS & INCIDENT TREND ANALYSIS

The annual trend analysis reveals a complex but ultimately escalating pattern of cybersecurity incidents and associated financial losses over the decade. While incident counts exhibit moderate variance year-on-year, financial losses display a pronounced upward trajectory post-2019, reflecting both increasing attack sophistication and growing digital asset values at risk.

Year	Incidents	Financial Loss ($M)	Avg. Loss/Incident ($M)	Trend
2015	277	$14,510	$52.4	Baseline
2016	285	$13,947	$48.9	Stable
2017	319	$16,262	$51.0	↑ Peak
2018	310	$14,720	$47.5	Correction

Year	Incidents	Financial Loss ($M)	Avg. Loss/Incident ($M)	Trend
2019	263	$13,135	$49.9	↓ Low
2020	315	$15,768	$50.1	↑ Recovery
2021	299	$15,873	$53.1	Elevated
2022	318	$15,871	$49.9	Sustained
2023	315	$15,958	$50.7	↑ High
2024	299	$15,434	$51.6	Elevated

The 2020 surge — coinciding with the global expansion of remote work infrastructure — saw incidents rebound sharply to 315, with financial losses climbing to $15.77B. Post-2020 losses have remained consistently elevated above $15.4B annually, suggesting a structural rather than cyclical shift in the threat landscape.

COUNTRY-LEVEL THREAT EXPOSURE

Geospatial analysis reveals that cybersecurity threats are a genuinely global phenomenon, with no single country immune from significant exposure. The UK records the highest cumulative financial loss at $16.5B, followed closely by Germany ($15.8B) and Brazil ($15.8B). Remarkably, even the lowest-ranked nation — China at $13.7B — sustains losses that would constitute a major economic event in most sectors.

Rank	Country	Total Loss ($M)	Share of Global	Incidents
1	United Kingdom	$16,503	10.9%	~300
2	Germany	$15,793	10.4%	~300
3	Brazil	$15,783	10.4%	~300
4	Australia	$15,403	10.2%	~300

Rank	Country	Total Loss ($M)	Share of Global	Incidents
5	Japan	$15,197	10.0%	~300
6	France	$14,972	9.9%	~300
7	USA	$14,812	9.8%	~300
8	Russia	$14,735	9.7%	~300
9	India	$14,566	9.6%	~300
10	China	$13,714	9.1%	~300

KEY INSIGHT: *The narrow spread between the highest (UK: $16.5B) and lowest (China: $13.7B) loss figures — a difference of only 20% — confirms that cyber threats are uniformly distributed globally. No geography offers a 'safe harbour' from sophisticated threat actors.*

ATTACK VECTOR DISTRIBUTION

The dataset captures six distinct attack vectors, each representing a unique threat paradigm. The distribution is remarkably balanced, with no single attack type commanding a majority — a finding that underscores the adaptive nature of modern threat actors who deploy diverse methodologies to maximise penetration success.

Attack Type	Incidents	Share	Primary Impact	Avg Resolution
DDoS	531	17.7%	Service Disruption	~36 hrs
Phishing	529	17.6%	Credential Theft / Financial Fraud	~36 hrs
SQL Injection	503	16.8%	Data Exfiltration	~37 hrs
Ransomware	493	16.4%	Data Encryption / Extortion	~37 hrs
Malware	485	16.2%	System Compromise	~36 hrs

Attack Type	Incidents	Share	Primary Impact	Avg Resolution
Man-in-the-Middle	459	15.3%	Data Interception	~36 hrs

***KEY INSIGHT**: DDoS and phishing jointly account for 35.3% of all incidents. The near-equal distribution across all six vectors means organisations cannot prioritise defence against a single attack type — comprehensive, multi-vector protection is essential.*

TARGET INDUSTRY VULNERABILITY ANALYSIS

Sector-level analysis reveals that the Information Technology sector bears the highest absolute financial burden, accumulating $24.81B in losses — representing 16.4% of global losses. Banking follows as the second most targeted sector ($22.77B), reflecting the high-value nature of financial data and the maturity of attack toolkits specifically engineered for financial system exploitation.

Industry	Total Loss ($M)	Loss Share	Risk Profile
Information Technology	$24,810	16.4%	Critical — Highest exposure
Banking & Finance	$22,772	15.0%	Critical — High-value targeting
Government	$21,205	14.0%	High — National security risk
Retail	$21,120	13.9%	High — Consumer data at risk
Healthcare	$21,041	13.9%	High — Patient data & services
Telecommunications	$20,459	13.5%	Medium-High — Infrastructure
Education	$20,071	13.2%	Medium — Research & PII data

***KEY INSIGHT**: Despite IT and Banking leading in losses, the Education sector's $20B exposure — comparable to Telecommunications — signals dangerous under-investment in cybersecurity for institutions often holding vast volumes of research IP and student PII.*

DEFENCE MECHANISM ADOPTION

Analysis of defence mechanisms deployed across the 3,000 incidents reveals a broadly distributed adoption pattern. Antivirus software leads with 628 deployments (20.9%), followed by VPN solutions (612, 20.4%), Encryption (592, 19.7%), Firewalls (585, 19.5%), and AI-based Detection (583, 19.4%). The near-parity across all five mechanisms is a structural observation of significant concern.

Defence Mechanism	Deployments	Frequency Share	Maturity Level	Effectiveness
Antivirus Software	628	20.9%	Legacy	Moderate — reactive
VPN (Virtual Private Network)	612	20.4%	Established	Moderate — perimeter
Encryption	592	19.7%	Established	High — data-at-rest/transit
Firewall	585	19.5%	Legacy	Moderate — perimeter
AI-based Detection	583	19.4%	Emerging	High — proactive/adaptive

KEY INSIGHT: *AI-based Detection — the most adaptive and proactive mechanism — is the least adopted at 19.4% despite its proven effectiveness against zero-day and novel attack patterns. Organisations relying predominantly on legacy tools (antivirus + firewall) remain structurally exposed to advanced persistent threats.*

INCIDENT RESOLUTION TIME ANALYSIS

The average incident resolution time across the dataset stands at 36.48 hours — approximately 1.5 days from detection to full containment. Resolution times range from a minimum of 1 hour to a maximum of 72 hours, with a median of 37 hours, indicating a right-skewed distribution where complex incidents significantly extend the tail.

Metric	Value	Implication
Minimum Resolution Time	1 hour	Best-case rapid containment scenarios
Average (Mean) Resolution Time	36.48 hours	Operational baseline for planning
Median Resolution Time	37 hours	Half of the incidents were resolved within 37 hrs
75th Percentile	55 hours	25% of incidents exceed 55 hours
Maximum Resolution Time	72 hours	Worst-case: 3-day full exposure window
Standard Deviation	20.57 hours	High variance — inconsistent response

A 36-hour average resolution window represents a critical vulnerability window during which attackers retain system access, potentially exfiltrating data, deploying secondary payloads, or establishing persistent backdoors. Industry benchmarks for mature security operations centres (SOCs) target sub-4-hour containment for critical incidents, suggesting the dataset average lags best practice by a factor of 9x.

***KEY INSIGHT**: The 20.57-hour standard deviation reveals extreme inconsistency in response capability. Organisations in the 75th percentile — taking 55+ hours — face disproportionately higher data loss and regulatory exposure. Automated response playbooks could reduce this variance by 40–60%.*

STRATEGIC RECOMMENDATIONS

Based on the analytical findings, the following strategic recommendations are presented for security leadership, risk officers, and technology decision-makers. These are prioritised by impact potential and urgency.

#	Recommendation	Priority
R1	Accelerate AI-Based Threat Detection	**CRITICAL**
R2	Implement Zero Trust Architecture (ZTA)	**CRITICAL**
R3	Reduce Incident Resolution Time via Automation	**HIGH**
R4	Establish Comprehensive Vulnerability Management	**HIGH**
R5	Prioritise IT & Banking Sector Security Investment	**MEDIUM**
R6	Develop Cross-Nation Cybersecurity Intelligence Sharing	**MEDIUM**

R1 — Accelerate AI-Based Threat Detection *(CRITICAL)* AI-based detection adoption at 19.4% is insufficient given the prevalence of zero-day exploits (26.2% of vulnerabilities). Organisations should transition from reactive, signature-based tools to ML-driven behavioural analytics platforms capable of detecting novel attack patterns in real-time. Target: 40%+ AI detection coverage by 2027.

R2 — Implement Zero Trust Architecture (ZTA) *(CRITICAL)* The 25.1% insider threat rate demands elimination of implicit trust. ZTA implementation — encompassing micro-segmentation, least-privilege access, and continuous authentication — is essential. Begin with identity-critical systems and financial data repositories, expanding to full network coverage within 18 months.

R3 — Reduce Incident Resolution Time via Automation *(HIGH)* The 36.48-hour average resolution time (vs. 4-hour SOC benchmark) represents an unacceptable exposure window. Deploy Security Orchestration, Automation and Response (SOAR) platforms with pre-

approved playbooks for common attack vectors. Target: sub-8-hour resolution for 80% of incidents within 12 months.

R4 — Establish Comprehensive Vulnerability Management *(HIGH)* With unpatched software (24.6%) and weak passwords (24.3%) each accounting for a quarter of exploited vulnerabilities, automated patch management and enterprise-wide MFA implementation offer high-ROI quick wins. Mandatory MFA + monthly patch cycles could eliminate ~50% of exploitable attack surfaces.

R5 — Prioritise IT & Banking Sector Security Investment *(MEDIUM)* IT ($24.81B loss) and Banking ($22.77B loss) sectors require dedicated threat-specific security programmes. IT organisations should focus on supply-chain security and API hardening; banking institutions should deploy fraud AI and real-time transaction monitoring as a baseline capability.

R6 — Develop Cross-Nation Cybersecurity Intelligence Sharing *(MEDIUM)* The uniform geographic loss distribution (range: $13.7B–$16.5B across 10 nations) confirms cyber threats transcend borders. Formalised bilateral and multilateral threat intelligence sharing frameworks — modelled on the ISAC network — would enable faster attribution and coordinated response to nation-state attacks.

CONCLUSION

This case study has systematically examined 3,000 cybersecurity incidents spanning a decade and 10 nations, surfacing a set of consistent, cross-cutting findings that demand urgent strategic attention from security and business leadership alike.

The global cybersecurity threat landscape is characterised by four defining features: **universality** — no geography, industry, or organisation size is immune; **diversity** — threat actors, attack vectors, and exploited vulnerabilities are broadly distributed, defying single-solution responses; **persistence** — with losses remaining above $13B annually even in lower-incident years; and **inadequacy of current defences** — evidenced by a 36.48-hour average resolution time and continued dominance of legacy security tools.

The Power BI dashboard analysis confirms that the threat environment has fundamentally shifted from opportunistic, volume-based attacks to sophisticated, targeted campaigns engineered to maximise financial and operational impact. The emergence of nation-state actors as the largest identifiable threat source (26.5%), combined with zero-day exploits as the leading vulnerability type (26.2%), signals an arms race that conventional security architectures are ill-equipped to win.

THE PATH AHEAD REQUIRES A DECISIVE SHIFT: FROM REACTIVE, PERIMETER-FOCUSED SECURITY TO PROACTIVE, INTELLIGENCE-LED, ZERO TRUST FRAMEWORKS ENHANCED BY AI. ORGANISATIONS THAT INVEST IN THESE CAPABILITIES NOW WILL NOT ONLY LESSEN THEIR FINANCIAL RISKS BUT ALSO BUILD THE INSTITUTIONAL RESILIENCE NEEDED TO OPERATE CONFIDENTLY IN AN INCREASINGLY HOSTILE DIGITAL LANDSCAPE.